

ISE 555: Unit 1 Assignment

Fall 2026

For this assignment, please complete at least **3 out of the first 5 problems**. The writing assignment (Problem 6) is mandatory. A PDF file containing all your solutions as well as the writing assignment must be submitted on Brightspace by 4:00pm on Monday, September 21st. Make sure you have access to your submission during class on the day that the assignment is due. This will be helpful when presenting your solutions to the class in person.

Please type your solutions. Suggest to use LaTeX (free online version is available <https://overleaf.com>)

1. A company manufactures an assembly consisting of a frame, a shaft, and a ball bearing. The company manufactures the shafts and the frames, but purchases the ball bearings from a ball bearing manufacturer. Each shaft must be processed on a forging machine, a lathe, and a grinder. These operations require 0.6 hour, 0.3 hour, and 0.4 hour per shaft, respectively. Each frame requires 0.8 hour on a forging machine, 0.2 hour on a drilling machine, 0.3 hour on a milling machine, and 0.6 hour on a grinder. The company has 5 lathes, 10 grinders, 20 forging machines, 3 drillers, and 6 millers. Assume that each machine operates a maximum of 4500 hours per year. Formulate the problem of finding the maximum number of assembled components that can be produced as a linear program.
2. An agricultural mill manufactures feed for chickens. This is done by mixing three main ingredients: (i) corn, (ii) limestone, and (iii) alfalfa. The mixing is done to ensure that the resulting feed meets certain levels for the different nutrients of (a) protein, (b) calcium, (c) carbohydrates, and (d) vitamins. We need a net amount of N bushels of the final feed mix. Let the unit cost of each ingredient be \$12/bushel for corn, \$5/bushel for limestone, and \$8/bushel for alfalfa. The corn contributes $\{c_{11}, c_{12}, c_{13}, c_{14}\}$ units of protein, calcium, carbohydrates, and vitamins per bushel, respectively. Similarly, the contribution of limestone are $\{c_{21}, c_{22}, c_{23}, c_{24}\}$ and those from the alfalfa are $\{c_{31}, c_{32}, c_{33}, c_{34}\}$. The feed is required to have a mix of at least $\{P_1, P_2, P_3, P_4\}$ units per bushel of each nutrient and no more than $\{u_1, u_2, u_3, u_4\}$ units per bushel of each nutrient.

u_4 units per bushel of each nutrient. Write a linear program to determine the optimal amount to order to minimize the cost and meet the need.

3. Suppose it takes a hen two weeks to lay 12 eggs for sale or to hatch 4 existing eggs into chicks. You are tasked to determine the best laying and hatching program if at the end of the fourth period all hens and chicks accumulated during the period are sold at \$2 apiece and eggs at \$0.2 apiece. Formulate the problem as a linear program under the following three scenarios:
 - (a) An initial inventory of 100 hens and 100 eggs
 - (b) An initial inventory of 100 hens and zero eggs
 - (c) An initial inventory as in (b) but also a final inventory of 100 hens and zero eggs
4. A government planning agency wishes to determine the purchasing sources for fuel for use by n depots from among m bidders. Suppose that the maximum quantity offered by bidder i is a_i gallons and that the demand of depot j is b_j gallons. Let c_{ij} be the unit delivery cost for bidder i to the j th depot.
 - (a) Formulate the problem of minimizing the total purchase cost as linear program
 - (b) Suppose that a discount in the unit delivery cost is offered by bidder i if the ordered quantity exceeds the level a_i . Modify the model in part (a) to incorporate this change.
5. Consider the production scheduling problem of the perfume Polly. The manufacturer of the perfume must plan production for the first four months of the year and anticipates a demand of 4000, 5000, 6000, and 4500 gallons in January, February, March, and April, respectively. At the beginning of the year the company has an inventory of 2000 gallons. The company is planning on issuing a new and improved perfume called Pollygone in May, so that all Polly produced must be sold by the end of April. Assume that the production cost for January and February is \$5 per gallon and this will rise to \$5.5 per gallon in March and April. The company can hold any amount produced in a certain month over to the next month at an inventory cost of \$1 per unit. Formulate a linear programming model that will minimize the costs

incurred in meeting the demand for Polly in the period January through April. Assume for simplicity that any amount produced in a given month may be used to fulfill demand for that month.

6. WRITING ASSIGNMENT:

You have been called to appear as an expert witness before the congressional committee that is reviewing the new budget of the US Department of Energy. In the past, the Department has provided a substantial amount of financial support for the development of optimization theory, algorithms, and software. Congressman Blank, a member of the committee, was never very good at mathematics and is hostile to this type of research. He has made recent headlines by reading off some of the titles of the more esoteric publications in this area. You are asked to prepare a statement (no more than 400 words) that explains why optimization work is important to include in the budget for the Department of Energy. [NOTE: This problem statement was 118 words long]